A Graph Contrastive Learning Framework with Multi-View Adaptation for Robust Collaborative Filtering

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Graph Notations & Neighborhood Definitions

One-Hop Neighborhoods on Bipartite Graph $G = (\mathcal{U} \cup \mathcal{I}, \mathcal{E})$

Let $\mathcal{U}$ denote the set of users ($|\mathcal{U}| = m$) and $\mathcal{I}$ denote the set of items ($|\mathcal{I}| = n$):

- $\mathcal{N}_G(u) = \{i \in \mathcal{I} \mid (u, i) \in \mathcal{E}\}$: Direct items interacted with by user u [1].
- $|\mathcal{N}_G(u)|$: Degree of user node u (interaction activity) [1].
- $\mathcal{N}_G(i) = \{u \in \mathcal{U} \mid (u, i) \in \mathcal{E}\}$: Direct users who interacted with item i [1].
- $|\mathcal{N}_G(i)|$: Degree of item node i (item popularity) [1].

Degree Normalization Factor & Matrix Form

$$\frac{1}{\sqrt{|\mathcal{N}_G(u)| \cdot |\mathcal{N}_G(i)|}} \iff \mathbf{D}_G^{-1/2} \mathbf{E}_G \mathbf{D}_G^{-1/2}, \qquad \mathbf{E}_G = \begin{pmatrix} \mathbf{0} & \mathbf{R} \\ \mathbf{R}^\top & \mathbf{0} \end{pmatrix}$$

- $\mathbf{R} \in \{0, 1\}^{m \times n}$ is the interaction matrix; $\mathbf{D}_G$ is the diagonal degree matrix with $(\mathbf{D}_G)_{xx} = |\mathcal{N}_G(x)|$ [2].
- Mitigates popularity bias and prevents high-degree hubs from dominating representations [1].
- Normalizes message passing across layers and stabilizes trust-score initialization [1].

Motivation & High-Order Collaborative Graph

Core Challenges in CF

- High interaction sparsity and noisy implicit feedback [1].
- Standard GNNs suffer from noise propagation and over-smoothing [1].

High-Order Collaborative Graph ($\tilde{G}$)

- Models user-user and item-item similarity via Jaccard index [1]:

$$JSC_{x,y} = \frac{|\mathcal{N}_G(x) \cap \mathcal{N}_G(y)|}{|\mathcal{N}_G(x) \cup \mathcal{N}_G(y)|}$$

- Retains edge if $JSC_{x,y} \geq \beta$ or if $y \in \text{top-}S(x)$, with $\beta = 0.8$ and $S = 5$ [1].
- Top-S guarantees every node at least S neighbors, which matters on sparse data [1].

Denoised Augmented Graph ($\overline{G}$)

1-Hop Neighborhood Aggregation

- Node vectors from the one-hop neighbors' initial embeddings [1]:

$$z_u^{G,1} = \sum_{i \in \mathcal{N}_G(u)} \frac{1}{\sqrt{|\mathcal{N}_G(u)| \times |\mathcal{N}_G(i)|}} z_i^{G,0}$$

$$z_i^{G,1} = \sum_{u \in \mathcal{N}_G(i)} \frac{1}{\sqrt{|\mathcal{N}_G(i)| \times |\mathcal{N}_G(u)|}} z_u^{G,0}$$

- Equivalently, in matrix form [2]:

$$\mathbf{Z}^{G,1} = \mathbf{D}_G^{-1/2} \mathbf{E}_G \mathbf{D}_G^{-1/2} \mathbf{Z}^{G,0}$$

- The symmetric normalization factor is used to balance the influence of nodes with different degrees, preventing high-degree nodes from dominating the aggregation [1].

Trust Score & Thresholding

- Normalized trust score in range $[0, 1]$ [1]:

$$T_{u,i} = \frac{1}{2}\left(\text{sim}(z_u^{G,1}, z_i^{G,1}) + 1\right)$$

- Hard/soft thresholding on interaction weights [1]:

$$\bar{r}_{u,i} = \begin{cases} T_{u,i}, & \text{if } T_{u,i} > \theta \\ 0, & \text{if } T_{u,i} \leq \theta \end{cases}$$

Multi-Graph Convolutions & Adaptive Fusion

1. **Multi-Graph Convolutional Propagation** (LightGCN-style matrix form [2])
   - **User-Item Graph (G)**: $\mathbf{Z}^{G,l} = \mathbf{D}_G^{-1/2}\,\mathbf{E}_G\,\mathbf{D}_G^{-1/2}\,\mathbf{Z}^{G,l-1}$ [1]
   - **High-Order Graph ($\tilde{G}$)**: $\mathbf{Z}^{\tilde{G},l} = \mathbf{D}_{\tilde{G}}^{-1}\,\mathbf{A}_{\tilde{G}}\,\mathbf{Z}^{\tilde{G},l-1}$ [1]
   - **Denoised Graph ($\overline{G}$)**: $\mathbf{Z}^{\overline{G},l} = \mathbf{D}_{\overline{G}}^{-1}\,\mathbf{E}_{\overline{G}}\,\mathbf{Z}^{\overline{G},l-1}, \quad \mathbf{E}_{\overline{G}} = \begin{pmatrix} \mathbf{0} & \bar{\mathbf{R}} \\ \bar{\mathbf{R}}^\top & \mathbf{0} \end{pmatrix}$ [1]

Here $\mathbf{Z}^{\cdot,l} \in \mathbb{R}^{(m+n)\times d}$ stacks all user/item embeddings at layer l (row $z_x^{\cdot,l}$ for node x); $\mathbf{A}_{\tilde{G}}$ is the unweighted adjacency of $\tilde{G}$ with $(\mathbf{D}_{\tilde{G}})_{xx} = |\mathcal{N}_{\tilde{G}}(x)|$; $\bar{\mathbf{R}} = [\bar{r}_{u,i}]$ and $\mathbf{D}_{\overline{G}}$ is the diagonal matrix of row sums of $\mathbf{E}_{\overline{G}}$.

2. **Layer-wise Dynamic Fusion**
   - Balances bipartite and high-order collaborative signals [1]:

$$z_x^l = z_x^{G,l} + \eta_x z_x^{\tilde{G},l}, \quad \text{where} \quad \eta_x = \frac{\gamma}{\mathrm{sim}(z_x^{G,l}, z_x^{\tilde{G},l}) \cdot d_x + l}$$

   - Normalized node activity: $d_x = \log(|\mathcal{N}_G(x)|) / \left(\frac{1}{|V|}\sum_{v\in V}\log(|\mathcal{N}_G(v)|)\right)$ [1].
   - Final layer readout via mean pooling across all L layers: $e_u = \frac{1}{L+1}\sum_{l=0}^{L} z_u^l$ [1].

Loss Functions (Part 1): Ranking & Multi-View Contrast

1. Bayesian Personalized Ranking Loss ($\mathcal{L}_{\mathrm{BPR}}$)

Directly optimizes pairwise preference margins between interacted positive items and unobserved negative items [1]:

$$\mathcal{L}_{\mathrm{BPR}} = \sum_{(u,i,j)\in\mathcal{O}} -\log \sigma(\hat{y}_{u,i} - \hat{y}_{u,j}), \quad \text{where } \hat{y}_{u,i} = e_u^\top e_i$$

Here, $\mathcal{O} = \{(u,i,j) \mid (u,i) \in \mathcal{E}, (u,j) \notin \mathcal{E}\}$ and $\sigma(\cdot)$ denotes the sigmoid activation function [1].

2. Multi-View Contrastive Loss ($\mathcal{L}_{\mathrm{CL}}$)

Aligns the unified fused view (e_v) with high-order collaborative ($e_v^{\tilde{G}}$) and denoised ($e_v^{\overline{G}}$) views [1]:

$$\mathcal{L}_{\mathrm{CL}}^{\tilde{G}} = \sum_{v\in\mathcal{B}} -\log \frac{\exp\left(\mathrm{sim}(e_v, e_v^{\tilde{G}})/\tau\right)}{\sum_{x\in\mathcal{B}} \exp\left(\mathrm{sim}(e_v, e_x^{\tilde{G}})/\tau\right)}, \quad \mathcal{L}_{\mathrm{CL}}^{\overline{G}} = \sum_{v\in\mathcal{B}} -\log \frac{\exp\left(\mathrm{sim}(e_v, e_v^{\overline{G}})/\tau\right)}{\sum_{x\in\mathcal{B}} \exp\left(\mathrm{sim}(e_v, e_x^{\overline{G}})/\tau\right)}$$

$$\mathcal{L}_{\mathrm{CL}} = \mathcal{L}_{\mathrm{CL}}^{\overline{G}} + \alpha \mathcal{L}_{\mathrm{CL}}^{\tilde{G}}$$

where $\mathcal{B}$ denotes the mini-batch and τ is the temperature hyperparameter [1].

Loss Functions (Part 2): Cross-Layer & Uniformity

3. Cross-Layer Contrastive Loss ($\mathcal{L}_{\text{layer}}$)

Aligns final fused embeddings e_v with intermediate representations $z_v^{\lfloor L/2 \rfloor}$ to prevent over-smoothing [1]:

$$\mathcal{L}_{\text{layer}} = \sum_{v \in \mathcal{B}} -\log \frac{\exp\left(\text{sim}(e_v, z_v^{\lfloor L/2 \rfloor})/\tau\right)}{\sum_{x \in \mathcal{B}} \exp\left(\text{sim}(e_v, z_x^{\lfloor L/2 \rfloor})/\tau\right)}$$

Forces deeper graph convolution layers to preserve informative low-layer semantics [1].

4. Hypersphere Uniformity Loss ($\mathcal{L}_{\text{UNIF}}$)

Enforces normalized representations $f(\cdot)$ to spread evenly across the unit hypersphere, preventing representation collapse [1]:

$$\mathcal{L}_{\text{unif}}^{u} = \log \mathbb{E}_{u,u' \sim \text{Uniq}(P_{\text{user}})} \left[e^{-2\|f(u)-f(u')\|_2^2} \right], \quad \mathcal{L}_{\text{unif}}^{i} = \log \mathbb{E}_{i,i' \sim \text{Uniq}(P_{\text{item}})} \left[e^{-2\|f(i)-f(i')\|_2^2} \right]$$

$$\mathcal{L}_{\text{UNIF}} = \frac{1}{2}\left(\mathcal{L}_{\text{unif}}^{u} + \mathcal{L}_{\text{unif}}^{i}\right)$$

Final Multi-Task Objective Function

MAGCL unifies ranking, multi-view alignment, hierarchical regularization, and parameter decay into a single joint objective [1]:

Overall Joint Objective

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{BPR}} + \lambda_1 \mathcal{L}_{\text{CL}} + \lambda_2 \mathcal{L}_{\text{layer}} + \lambda_3 \mathcal{L}_{\text{UNIF}} + \lambda_4 \|\Theta\|_2$$

- $\mathcal{L}_{\text{BPR}}$: Primary recommendation objective optimizing relative pairwise ranking margins [1].
- $\lambda_1 \mathcal{L}_{\text{CL}}$: Multi-view contrastive loss regularizing against noisy and high-order graphs [1].
- $\lambda_2 \mathcal{L}_{\text{layer}}$: Cross-layer contrastive loss suppressing over-smoothing across GNN depth [1].
- $\lambda_3 \mathcal{L}_{\text{UNIF}}$: Hypersphere uniformity penalty maximizing embedding diversity [1].
- $\lambda_4 \|\Theta\|_2$: L_2-norm regularization on trainable parameters Θ to prevent overfitting [1].

Baseline Results (LightGCN) [1 seed]

LightGCN Benchmark Comparison on MovieLens-1M

Metric	Paper Reported [1]	Our Run (1 seed)	Relative Gap (%)
Recall@10	0.1876	0.1790	4.58
NDCG@10	0.2514	0.2466	1.91
Recall@20	0.2796	0.2674	4.36
NDCG@20	0.2620	0.2558	2.37
Recall@50	0.4469	0.4313	3.49
NDCG@50	0.3091	0.3017	2.39

- Single-seed reproduction using the standard RecBole LightGCN model [2].
- Paper values are the LightGCN baseline reported by Jiang et al. (Table 2) [1].
- Relative gap $=$ (Paper $-$ Ours)/Paper $\times$ 100.

MAGCL Implementation Results

Our MAGCL Implementation vs. Paper on MovieLens-1M

Metric	Paper (Table 2) [1]	Our Run (1 seed, test)	Relative Gap (%)
Recall@10	0.2088	0.1976	5.36
NDCG@10	0.2674	0.2560	4.26
Recall@20	0.3084	0.2985	3.21
NDCG@20	0.2819	0.2722	3.44
Recall@50	0.4780	0.4673	2.24
NDCG@50	0.3315	0.3214	3.05

- Our full implementation of MAGCL evaluated on the test split; paper values are MAGCL's results in Table 2 [1].
- Relative gap $=$ (Paper $-$ Ours)/Paper $\times$ 100; all six metrics are within about 5.4% of the paper.
- Our MAGCL run improves over our LightGCN run on every metric [2].

Current Work & Future Work

Current Work

- Full implementation and pipeline reproduction of MAGCL modules.
- Systematic tuning across 5 key hyperparameters [1]:
  - λ_1: Multi-view contrastive loss weight
  - λ_2: Cross-layer contrastive loss weight
  - λ_3: Uniformity regularization weight
  - λ_4: L_2 regularization / weight decay
  - θ: Trust-score denoising cutoff threshold

Future Work

- Implement and benchmark 4 additional baselines:
  - SGL (Self-Supervised Graph Learning)
  - SimGCL (Simple GCL)
  - AdaMCL (Adaptive Multi-View CL) [1]
  - ProtoAU (Prototypical Alignment/Uniformity) [1]
- Multi-seed evaluation across all the baselines.

References



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